

DDA3020 HW1

1.1. Weighted Linear Regression

(a)

Define

$$\begin{aligned}\tilde{x}_i &= \begin{bmatrix} x_i \\ 1 \end{bmatrix} \in \mathbb{R}^{d+1}, \\ \tilde{X} &= \begin{bmatrix} \tilde{x}_1^T \\ \vdots \\ \tilde{x}_N^T \end{bmatrix} \in \mathbb{R}^{N \times (d+1)}, \\ Y &= \begin{bmatrix} y_1^T \\ \vdots \\ y_N^T \end{bmatrix} \in \mathbb{R}^{N \times m}, \\ A &= \text{diag}(\alpha_1, \dots, \alpha_N). \\ \tilde{W} &= [W \quad b] \in \mathbb{R}^{m \times (d+1)}.\end{aligned}$$

Then

$$J(W, b) = \sum_{i=1}^N \alpha_i \|y_i - (Wx_i + b)\|^2 = \text{tr}\left((Y - \tilde{X}\tilde{W}^T)^T A (Y - \tilde{X}\tilde{W}^T)\right).$$

Derivative:

$$\frac{\partial J}{\partial \tilde{W}} = -2A(Y - \tilde{X}\tilde{W}^T)^T \tilde{X} = -2(Y^T A \tilde{X} - \tilde{W} \tilde{X}^T A \tilde{X}).$$

Setting it to zero:

$$\tilde{X}^T A \tilde{X} \tilde{W}^T = \tilde{X}^T A Y.$$

Hence the closed-form solution is

$$\tilde{W}^{*T} = (\tilde{X}^T A \tilde{X})^{-1} \tilde{X}^T A Y.$$

i.e.,

$$\tilde{W}^* = Y^T A \tilde{X} (\tilde{X}^T A \tilde{X})^{-1},$$

and we get W^* (the first d columns of $\tilde{W}^*$) and b^* (the last column of $\tilde{W}^*$).

(b)

Let $r_i = y_i - (Wx_i + b)$.

Then

$$J(W, b) = \sum_{i=1}^N \alpha_i r_i^T r_i.$$

Gradients:

$$\frac{\partial J}{\partial W} = -2 \sum_{i=1}^N \alpha_i r_i x_i^T,$$
$$\frac{\partial J}{\partial b} = -2 \sum_{i=1}^N \alpha_i r_i.$$

Gradient descent updates (learning rate η):

$$W^{(t+1)} = W^{(t)} - \eta \frac{\partial J}{\partial W} = W^{(t)} + 2\eta \sum_{i=1}^N \alpha_i (y_i - (W^{(t)} x_i + b^{(t)})) x_i^T,$$
$$b^{(t+1)} = b^{(t)} - \eta \frac{\partial J}{\partial b} = b^{(t)} + 2\eta \sum_{i=1}^N \alpha_i (y_i - (W^{(t)} x_i + b^{(t)})).$$

1.2. Mathematical Foundations: Optimization & Information Theory

(a)

Objective:

$$f_0(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 1)^2.$$

Gradient:

$$\nabla f_0 = \begin{bmatrix} 2(x_1 - 2) \\ 2(x_2 - 1) \end{bmatrix}.$$

Hessian:

$$\nabla^2 f_0 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$$

which is positive definite, so f_0 is convex.

The constraints

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0$$

are affine (linear) constraints, hence they define a convex feasible set.

Therefore the problem is a convex optimization problem.

Then we shall deal with KKT conditions:

Write constraints in the form $g_i(x) \leq 0$:

$$g_1(x) = x_1 + x_2 - 1 \leq 0, \quad g_2(x) = -x_1 \leq 0, \quad g_3(x) = -x_2 \leq 0.$$

Lagrangian:

$$L(x, \mu) = (x_1 - 2)^2 + (x_2 - 1)^2 + \mu_1(x_1 + x_2 - 1) - \mu_2 x_1 - \mu_3 x_2, \quad \mu_1, \mu_2, \mu_3 \geq 0.$$

KKT conditions

1. **Stationarity**

$$\nabla_x L = 0 \Rightarrow \begin{cases} 2(x_1 - 2) + \mu_1 - \mu_2 = 0, \\ 2(x_2 - 1) + \mu_1 - \mu_3 = 0. \end{cases}$$

2. Primal feasibility

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

3. Dual feasibility

$$\mu_1, \mu_2, \mu_3 \geq 0.$$

4. Complementary slackness

$$\mu_1(x_1 + x_2 - 1) = 0, \quad \mu_2 x_1 = 0, \quad \mu_3 x_2 = 0.$$

(b)

From stationarity, we have:

$$x_1 = 2 - \frac{\mu_1}{2}, \quad x_2 = 1 - \frac{\mu_1}{2}.$$

Given that $\mu_1(x_1 + x_2 - 1) = 0$, we have 2 cases here:

Case 1: $\mu_1 = 0$.

Then $x_1 = 2, x_2 = 1$, which violates the primal feasibility condition $x_1 + x_2 \leq 1$.

So $\mu_1 = 0$ is not possible.

Case 2: $\mu_1 > 0, x_1 + x_2 = 1$.

At the optimum we will have $x_1 > 0$, so $\mu_2 = 0$ by $\mu_2 x_1 = 0$.

Also, the closest point on the line $x_1 + x_2 = 1$ in the first quadrant occurs at $x_2 = 0$, so $x_2 = 0$ (given $\mu_3 x_2 = 0$).

Using stationarity condition:

$$2(1 - 2) + \mu_1 - \mu_2 = 0 \Rightarrow -2 + \mu_1 = 0 \Rightarrow \mu_1 = 2,$$

$$2(0 - 1) + \mu_1 - \mu_3 = 0 \Rightarrow -2 + 2 - \mu_3 = 0 \Rightarrow \mu_3 = 0.$$

All other KKT conditions are also satisfied, so

$$x^* = (1, 0).$$

(c)

(1)

$$D_{\text{KL}}(P \| Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)},$$

$$D_{\text{KL}}(Q \| P) = \sum_x Q(x) \log \frac{Q(x)}{P(x)}.$$

Difference:

$$\begin{aligned}
& D_{\text{KL}}(P\|Q) - D_{\text{KL}}(Q\|P) \\
&= \sum_x P(x) \log \frac{P(x)}{Q(x)} + \sum_x Q(x) \log \frac{P(x)}{Q(x)} \\
&= \sum_x (P(x) + Q(x)) \log \frac{P(x)}{Q(x)}.
\end{aligned}$$

In general, $\sum_x (P(x) + Q(x)) \log \frac{P(x)}{Q(x)} \neq 0$ (To support this argument, the next section gives a simple counterexample, i.e.,

$$\begin{aligned}
P(1) = \frac{1}{3}, P(2) = \frac{2}{3}; \quad Q(1) = \frac{1}{2}, Q(2) = \frac{1}{2}. \quad D_{\text{KL}}(P\|Q) = \frac{1}{3} \log \frac{2}{3} + \frac{2}{3} \log \frac{4}{3} \approx 0.05663, \\
D_{\text{KL}}(Q\|P) = \frac{1}{2} \log \frac{3}{2} + \frac{1}{2} \log \frac{3}{4} = \frac{1}{2} \log \frac{9}{8} \approx 0.05889, \text{ hence}
\end{aligned}$$

$$D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$$

(2)

Let the sample space be $\{1, 2\}$, with

$$P(1) = \frac{1}{3}, P(2) = \frac{2}{3}; \quad Q(1) = \frac{1}{2}, Q(2) = \frac{1}{2}.$$

Then

$$\begin{aligned}
D_{\text{KL}}(P\|Q) &= \frac{1}{3} \log \frac{2}{3} + \frac{2}{3} \log \frac{4}{3} \approx 0.05663, \\
D_{\text{KL}}(Q\|P) &= \frac{1}{2} \log \frac{3}{2} + \frac{1}{2} \log \frac{3}{4} = \frac{1}{2} \log \frac{9}{8} \approx 0.05889.
\end{aligned}$$

So $D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$.

1.3. Logistic Regression Properties

(a)

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Differentiate:

$$\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2}.$$

Also,

$$\sigma(z)(1 - \sigma(z)) = \frac{1}{1 + e^{-z}} \left(1 - \frac{1}{1 + e^{-z}}\right) = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \frac{e^{-z}}{(1 + e^{-z})^2}.$$

Hence,

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)).$$

(b)

For one sample,

$$J_i(w) = -\left(y_i \log h_i + (1 - y_i) \log(1 - h_i)\right), \quad h_i = \sigma(w^T x_i).$$

Using chain rule:

$$\frac{\partial J_i}{\partial w} = (h_i - y_i)x_i.$$

Summing it up:

$$\nabla_w J(w) = \sum_{i=1}^n (h_i - y_i)x_i.$$

Let $X \in \mathbb{R}^{n \times d}$ be the data matrix (rows are x_i^T), $h \in \mathbb{R}^n$ and $y \in \mathbb{R}^n$.

Then

$$\nabla_w J(w) = X^T(h - y).$$

(c)

Let $S = \text{diag}(h_i(1 - h_i))$ (so $S \succeq 0$).

Hessian:

$$H = \nabla_w^2 J(w) = X^T S X.$$

For any vector v ,

$$v^T H v = v^T X^T S X v = (Xv)^T S (Xv) = \sum_{i=1}^n s_i (x_i^T v)^2 \geq 0,$$

This is because each $s_i = h_i(1 - h_i) \geq 0$.

Thus H is PSD, and $J(w)$ is convex.

(d)

With L2 regularization:

$$J_{\text{reg}}(w) = J(w) + \frac{\lambda}{2} w^T w.$$

Gradient:

$$\nabla_w J_{\text{reg}}(w) = X^T(h - y) + \lambda w.$$

Gradient descent update:

$$w^{(t+1)} = w^{(t)} - \eta \left(X^T(h^{(t)} - y) + \lambda w^{(t)} \right).$$

(e)

For softmax regression,

$$\begin{aligned} P_{ik} &= \frac{e^{w_k^T x_i}}{\sum_{j=1}^K e^{w_j^T x_i}}, \\ L &= - \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log P_{ik} \\ &= - \sum_{i=1}^n \sum_{k=1}^K y_{ik} (w_k^T x_i - \log \sum_{j=1}^K e^{w_j^T x_i}) \end{aligned}$$

Since

$$\nabla_{w_k} \left(- \sum_{i=1}^n \sum_{j=1}^K y_{ij} w_j^T x_i \right) = - \sum_{i=1}^n y_{ik} x_i,$$

and

$$\begin{aligned} \nabla_{w_k} \left(\sum_{i=1}^n \sum_{j=1}^K y_{ij} \log \sum_{l=1}^K e^{w_l^T x_i} \right) \\ = \nabla_{w_k} \left(\sum_{i=1}^n \log \sum_{l=1}^K e^{w_l^T x_i} \right) \\ = \sum_{i=1}^n \frac{e^{w_k^T x_i}}{\sum_{j=1}^K e^{w_j^T x_i}} x_i \\ = \sum_{i=1}^n P_{ik} x_i, \end{aligned}$$

Summing the two equations, and we have the gradient:

$$\nabla_{w_k} L = \sum_{i=1}^n (P_{ik} - y_{ik}) x_i.$$

In matrix form (letting $p_k = [P_{1k}, \dots, P_{nk}]^T$ and $y_k = [y_{1k}, \dots, y_{nk}]^T$):

$$\nabla_{w_k} L = X^T (p_k - y_k).$$

2.1. Gradient Descent from Scratch

(a)

x1	0.0000713624 ≈ 0
x2	0.0000000000 ≈ 0
f(x)	5.092589940836e - 09 ≈ 0

(b)

(c)

eta	iterations
0.01	converged in 798 iterations
0.1	converged in 73 iterations
1.1	diverged by iter 10000

- Reason for divergence for eta = 1.1:

$$x_2^{(t+1)} = (1 - 10\eta) x_2^{(t)}$$

- For $\eta = 1.1$, we have $x_2^{(t+1)} = (1 - 11)x_2^{(t)} = -10x_2^{(t)}$. This means that the value of x_2 will grow exponentially in magnitude, leading to divergence.

2.2. Logistic Regression with Regularization

⚠ Warning

- $Seed = 1$
- Hyperparameters: $lr(\eta)=0.1$, $epochs=10000$

💡 Important

Aligned with lecture notes, cost functions for L1 and L2 regularization are defined as follows:

- $$L2 : \bar{J}(w) = J(w) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2$$
- $$L1 : \bar{J}(w) = J(w) + \frac{\lambda}{m} \sum_{j=1}^d |w_j|$$

(a)

Train acc = 0.9750, Test acc = 1.0000

(b) L2

λ	Train acc	Test acc	n(zero weights)
0.01	0.9750	1.0000	0
0.1	0.9750	1.0000	0
1.0	0.9500	0.9667	0

(c) L1

λ	Train acc	Test acc	n(zero weights)
0.01	0.9750	1.0000	0
0.1	0.9750	1.0000	0
1.0	0.9667	1.0000	4

(d)

Comparison:

- Model Performance:
 - Both L1 and L2 regularization with λ values of 0.01 and 0.1 achieved perfect test accuracy (1.0000) and high train accuracy (0.9750).

- However, at $\lambda = 1.0$, L2 regularization resulted in a decrease in both train and test accuracy (0.9500 and 0.9667), while L1 regularization maintained a higher test accuracy (1.0000) despite a slight drop in train accuracy (0.9667).
- Sparsity:
 - L1 regularization with $\lambda = 1.0$ resulted in 4 out of 12 weights being near-zero, indicating that L1 regularization promotes sparsity in the learned weights.
 - In contrast, L2 regularization did not produce any near-zero weights across all λ values, suggesting that L2 regularization does not encourage sparsity and rather tends to shrink weights.